

Triangularization and Duality of the Normalized IQC System

Let P be the primal input–output operator and let

$$\Psi = \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix}, \quad J_{z,w} = \begin{bmatrix} I_z & 0 \\ 0 & -I_w \end{bmatrix}, \quad J_{w,z} = \begin{bmatrix} I_w & 0 \\ 0 & -I_z \end{bmatrix}.$$

Assume that Ψ is boundedly and causally invertible. To use the triangularization of Yuan and Wu, assume additionally that Ψ_{22} has a bounded causal inverse. Finally, assume that

$$F_0 := \Psi_{21}P + \Psi_{22}$$

has a bounded causal inverse. This is the input–output counterpart of the current assumption in Lemma 1, $(\mathcal{D}_{\Theta}^{wz}\mathcal{D} + \mathcal{D}_{\Theta}^w)^{-1} \in \Pi_4$. Throughout, $\top$ denotes the causal dual of an input–output operator; $*$ is retained for realization operators.

Triangularize the factor and normalize the primal graph

Following Yuan and Wu, introduce the invertible signal transformation

$$L_{\Psi} := \begin{bmatrix} I & 0 \\ \Psi_{21} & \Psi_{22} \end{bmatrix}, \quad \begin{bmatrix} z \\ \tilde{w} \end{bmatrix} = L_{\Psi} \begin{bmatrix} z \\ w \end{bmatrix}.$$

Moving this transformation from the factor to the interconnection gives

$$\Psi_{\Delta} := \Psi L_{\Psi}^{-1} = \begin{bmatrix} \Psi_{11} - \Psi_{12}\Psi_{22}^{-1}\Psi_{21} & \Psi_{12}\Psi_{22}^{-1} \\ 0 & I \end{bmatrix}.$$

Hence the triangular form is obtained by an invertible change of internal signals; it is not an additional restriction on the original factor. Moreover,

$$\Psi \begin{bmatrix} z \\ w \end{bmatrix} = \Psi_{\Delta} \begin{bmatrix} z \\ \tilde{w} \end{bmatrix}.$$

On the graph $z = Pw$, the same transformation gives

$$L_{\Psi} \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} P \\ F_0 \end{bmatrix} w = \begin{bmatrix} PF_0^{-1} \\ I \end{bmatrix} F_0 w.$$

Consequently,

$$\Psi \begin{bmatrix} P \\ I \end{bmatrix} w = \Psi_{\Delta} \begin{bmatrix} PF_0^{-1} \\ I \end{bmatrix} F_0 w = \begin{bmatrix} G_0 \\ I \end{bmatrix} F_0 w,$$

where

$$G_0 = (\Psi_{11}P + \Psi_{12})(\Psi_{21}P + \Psi_{22})^{-1}.$$

Since $\tilde{w} = F_0 w$, the normalized system is

$$\tilde{z} = G_0 \tilde{w}. \tag{21}$$

The primal IQC is now an ordinary gain condition:

$$\begin{aligned}\mathcal{I}_T(w) &:= \left\langle p_T \Psi \begin{bmatrix} P \\ I \end{bmatrix} w, J_{z,w} p_T \Psi \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle_{L_{\mathbf{H}}} \\ &= \|p_T G_0 F_0 w\|^2 - \|p_T F_0 w\|^2.\end{aligned}$$

If $\|G_0\| \leq \rho < 1$, the bounded causal invertibility of F_0 assumed at the start gives

$$\mathcal{I}_T(w) \leq -\frac{1-\rho^2}{\|F_0^{-1}\|^2} \|p_T w\|^2.$$

In a state-space representation, L_Ψ induces the corresponding invertible congruence transformation between the original filtered realization and the normalized realization.

Obtain the paired dual transformation

Define

$$\hat{J} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad \mathbf{D}(\Psi) := (\hat{J}^* \Psi^\top \hat{J})^{-1}.$$

Calculate the dual filter directly. Since

$$\hat{J}^* \Psi^\top \hat{J} = \begin{bmatrix} \Psi_{22}^\top & -\Psi_{12}^\top \\ -\Psi_{21}^\top & \Psi_{11}^\top \end{bmatrix},$$

and

$$(\Psi_{\Delta,11})^\top = \Psi_{11}^\top - \Psi_{21}^\top (\Psi_{22}^\top)^{-1} \Psi_{12}^\top,$$

the block inverse formula gives

$$\mathbf{D}(\Psi) = \begin{bmatrix} (\Psi_{22}^\top)^{-1} + (\Psi_{22}^\top)^{-1} \Psi_{12}^\top ((\Psi_{\Delta,11})^\top)^{-1} \Psi_{21}^\top (\Psi_{22}^\top)^{-1} & (\Psi_{22}^\top)^{-1} \Psi_{12}^\top ((\Psi_{\Delta,11})^\top)^{-1} \\ ((\Psi_{\Delta,11})^\top)^{-1} \Psi_{21}^\top (\Psi_{22}^\top)^{-1} & ((\Psi_{\Delta,11})^\top)^{-1} \end{bmatrix}.$$

The desired paired factorization can now be read off directly:

$$\underline{\Psi}_\Delta := \begin{bmatrix} (\Psi_{22}^\top)^{-1} & (\Psi_{22}^\top)^{-1} \Psi_{12}^\top \\ 0 & I \end{bmatrix}, \quad \underline{L}_\Psi := \begin{bmatrix} I & 0 \\ \mathbf{D}(\Psi)_{21} & \mathbf{D}(\Psi)_{22} \end{bmatrix}.$$

Normalize the dual graph

Write $S := \Psi_{\Delta,11}$. The lower block of $\underline{L}_\Psi$ defines the dual normalizer

$$\underline{\Gamma}_0 := \mathbf{D}(\Psi)_{21} P^\top + \mathbf{D}(\Psi)_{22} = S^{-\top} \left(I + \Psi_{21}^\top \Psi_{22}^{-\top} P^\top \right).$$

Equivalently,

$$\underline{\Gamma}_0^\top = (I + P \Psi_{22}^{-1} \Psi_{21}) S^{-1}.$$

Because $F_0 = \Psi_{22}(I + \Psi_{22}^{-1} \Psi_{21} P)$ is invertible, the push-through rule shows that $I + P \Psi_{22}^{-1} \Psi_{21}$ is invertible. Hence $\underline{\Gamma}_0$ is invertible, with

$$\underline{\Gamma}_0^{-\top} = S (I + P \Psi_{22}^{-1} \Psi_{21})^{-1}.$$

On the dual graph $\underline{z} = P^\top \underline{w}$, the transformation $\underline{L}_\Psi$ now gives

$$\underline{L}_\Psi \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} P^\top \\ \underline{\Gamma}_0 \end{bmatrix} \underline{w} = \begin{bmatrix} P^\top \underline{\Gamma}_0^{-1} \\ I \end{bmatrix} \underline{\Gamma}_0 \underline{w}.$$

Consequently,

$$\begin{aligned} \mathbf{D}(\Psi) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} &= \underline{\Psi}_\Delta \begin{bmatrix} P^\top \underline{\Gamma}_0^{-1} \\ I \end{bmatrix} \underline{\Gamma}_0 \underline{w} \\ &= \begin{bmatrix} \underline{G}_0 \\ I \end{bmatrix} \underline{\Gamma}_0 \underline{w}, \end{aligned}$$

where

$$\underline{G}_0 = (\Psi_{22}^\top)^{-1} \left(P^\top \underline{\Gamma}_0^{-1} + \Psi_{12}^\top \right).$$

The same push-through identity gives

$$(I + P\Psi_{22}^{-1}\Psi_{21})^{-1}P\Psi_{22}^{-1} = P(\Psi_{21}P + \Psi_{22})^{-1} = PF_0^{-1}.$$

It therefore identifies the normalized dual operator in three lines:

$$\begin{aligned} \underline{G}_0^\top &= \underline{\Gamma}_0^{-\top} P\Psi_{22}^{-1} + \Psi_{12}\Psi_{22}^{-1} \\ &= (SP + \Psi_{12}\Psi_{22}^{-1}F_0) F_0^{-1} \\ &= (\Psi_{11}P + \Psi_{12})F_0^{-1} = G_0. \end{aligned}$$

Hence, with $\tilde{\underline{w}} := \underline{\Gamma}_0 \underline{w}$, the normalized dual system is

$$\tilde{\underline{z}} = G_0^\top \tilde{\underline{w}}.$$

Equivalently,

$$G_0^\top = (P^\top \Psi_{21}^\top + \Psi_{22}^\top)^{-1} (P^\top \Psi_{11}^\top + \Psi_{12}^\top),$$

where the first factor is $(F_0^\top)^{-1} = (F_0^{-1})^\top$.

The dual IQC is now the same ordinary gain condition:

$$\begin{aligned} \underline{\mathcal{I}}_T(\underline{w}) &:= \left\langle p_T \mathbf{D}(\Psi) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w}, J_{w,z} p_T \mathbf{D}(\Psi) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{L_H} \\ &= \|p_T G_0^\top \underline{\Gamma}_0 \underline{w}\|^2 - \|p_T \underline{\Gamma}_0 \underline{w}\|^2. \end{aligned}$$

Since $\|G_0^\top\| = \|G_0\| \leq \rho < 1$, bounded causal invertibility of $\underline{\Gamma}_0$ gives

$$\underline{\mathcal{I}}_T(\underline{w}) \leq -\frac{1-\rho^2}{\|\underline{\Gamma}_0^{-1}\|^2} \|p_T \underline{w}\|^2.$$

For the sign comparison, no strict upper bound on ρ is needed. For any common gain bound $\|G_0\| = \|G_0^\top\| \leq \rho$, define the pullback coefficients

$$\varepsilon_p := \frac{1-\rho^2}{\|F_0^{-1}\|^2}, \quad \varepsilon_d := \frac{1-\rho^2}{\|\underline{\Gamma}_0^{-1}\|^2}.$$

Because both normalizers are boundedly invertible, their inverse norms are finite and strictly positive. Consequently,

$$\text{sgn}(\varepsilon_p) = \text{sgn}(1-\rho^2) = \text{sgn}(\varepsilon_d).$$

In particular, $\varepsilon_p > 0$ if and only if $\varepsilon_d > 0$. A strict bound $\rho < 1$ is needed only to conclude that this common sign is positive, in which case the coefficients are valid strict IQC margins. Thus primal and dual strictness are equivalent, although the two positive margins need not have the same magnitude.